Exercise Sheet 6

The notation is that of Lecture 6: V_m has basis $v_k = f^k v_0$, $M[\mu]$ is a weight space, and $V \otimes W$ carries the tensor action. Modules are over $\mathbb{C}$. They are finite-dimensional except for $\mathbb{C}[x,y]$ in Exercise 6.1 and the modules in Exercise 6.7. Exercises 6.1–6.4 develop the lecture. Exercises 6.5–6.7 apply its methods to further modules.

Exercise 6.1 (Symmetric powers). Let $x = (1,0)^{\mathsf{T}}$ and $y = (0,1)^{\mathsf{T}}$ in the defining module $\mathbb{C}^2$. Identify $\mathrm{Sym}^m(\mathbb{C}^2)$ with the homogeneous polynomials of degree m in x and y. Let $X \in \mathfrak{sl}_2$ act on $\mathbb{C}[x,y]$ by the unique derivation $\rho(X)$ that agrees with X on x and y.

(i) Show that $\rho(e) = x\,\partial_y$, $\rho(f) = y\,\partial_x$ and $\rho(h) = x\,\partial_x - y\,\partial_y$. Prove that ρ is a representation preserving each $\mathrm{Sym}^m(\mathbb{C}^2)$.

(ii) Prove that $f^k x^m = \frac{m!}{(m-k)!}\, x^{m-k} y^k$ for $0 \le k \le m$.

(iii) Deduce an explicit isomorphism $V_m \to \mathrm{Sym}^m(\mathbb{C}^2)$.

Exercise 6.2 (Invariant bilinear forms on V_m). A bilinear form B on a module M is *invariant* if

$$B(xu, w) + B(u, xw) = 0 \qquad \text{for all } x \in \mathfrak{sl}_2,\ u, w \in M.$$

(i) Let B be invariant on V_m and put $b_0 = B(v_0, v_m)$. Prove that $B(v_k, v_l) = 0$ unless $k+l = m$, and that $B(v_k, v_{m-k}) = (-1)^k b_0$.

(ii) Prove conversely that these values define an invariant form for every $b_0 \in \mathbb{C}$. Deduce that the invariant forms on V_m form a one-dimensional space, and that each nonzero one is nondegenerate.

(iii) Prove that B is symmetric for even m and alternating for odd m.

(iv) For $m = 2$, use the adjoint realisation $v_0 = e$, $v_1 = -h$, $v_2 = -2f$ of Lecture 6 and the Killing form $K(x,y) = 4\,\mathrm{tr}(xy)$. Find b_0.

Exercise 6.3 (Singlet vectors). A *singlet* in a module M is a vector s with $xs = 0$ for all $x \in \mathfrak{sl}_2$.

(i) Prove that the singlets form a subspace of dimension $\dim M[0] - \dim M[2]$.

(ii) Deduce that $V_m \otimes V_n$ has a nonzero singlet only if $m = n$, and then a one-dimensional space of singlets.

(iii) Prove that

$$s_m = \sum_{k=0}^{m} (-1)^k\, v_k \otimes v_{m-k}$$

is a singlet in $V_m \otimes V_m$. Show that the flip $P(u \otimes w) = w \otimes u$ maps s_m to $(-1)^m s_m$.

(iv) Write out s_1 and s_2.

Exercise 6.4 (The Casimir operator on a tensor product). Let Ω be the Casimir operator of Lecture 6, for the form $B(x,y) = \mathrm{tr}(xy)$.

(i) For modules V and W, prove that $\Omega_{V\otimes W} = \Omega_V \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \Omega_W + 2T$, where

$$T = \frac{1}{2}\rho_V(h) \otimes \rho_W(h) + \rho_V(e) \otimes \rho_W(f) + \rho_V(f) \otimes \rho_W(e).$$

(ii) Prove that T acts on every submodule of $V_m \otimes V_n$ isomorphic to V_{m+n-2k} by the scalar

$$t_k = \frac{mn}{2} - k(m+n-k+1).$$

(iii) For $m = n = 1$, compute T on the basis $v_i \otimes v_j$ and show $T = P - \frac{1}{2}$, where P is the flip. Compare with (ii).

Optional: the half-integer parameter j. Put $m = 2j$. The eigenvalues of $\frac{1}{2}\rho(h)$ on V_m are $j, j-1, \dots, -j$, and $\dim V_m = 2j+1$. The scalar of $\frac{1}{2}\Omega$ is $j(j+1)$.

Exercise 6.5 (The tensor product $V_m \otimes V_1$). Let $m \geq 1$. Let $v_0, \dots, v_m$ and w_0, w_1 be the standard bases of V_m and V_1, with $v_k = 0$ for $k < 0$ or $k > m$.

(i) Prove that $f^k(a \otimes b) = \sum_{j=0}^{k} \binom{k}{j} f^{k-j}a \otimes f^j b$.

(ii) Show that $u = v_0 \otimes w_0$ and $x = v_1 \otimes w_0 - m\, v_0 \otimes w_1$ are highest vectors, and find their weights.

(iii) Prove that, for $k \geq 0$,

$$f^k u = v_k \otimes w_0 + k\, v_{k-1} \otimes w_1, \qquad f^k x = v_{k+1} \otimes w_0 + (k-m)\, v_k \otimes w_1.$$

(iv) Prove that the vectors $f^k u$ for $0 \leq k \leq m+1$ and $f^k x$ for $0 \leq k \leq m-1$ form a basis. Deduce $V_m \otimes V_1 = \iota(V_{m+1}) \oplus \iota'(V_{m-1})$ for explicit inclusions ι and ι'.

Exercise 6.6 (Symmetric and alternating squares of V_2). Let V be a module and P the flip of $V \otimes V$.

(i) Prove that P is an intertwiner. We use the tensor models $\mathrm{Sym}^2 V = \ker(P-1)$ and $\Lambda^2 V = \ker(P+1)$ inside $V \otimes V$. Deduce that these are submodules with $V \otimes V = \mathrm{Sym}^2 V \oplus \Lambda^2 V$.

(ii) For $V = V_2$, prove that $\mathrm{Sym}^2 V_2 \cong V_4 \oplus V_0$ and $\Lambda^2 V_2 \cong V_2$.

(iii) In the adjoint realisation $v_0 = e$, $v_1 = -h$, $v_2 = -2f$, show that the singlet s_2 of Exercise 6.3 equals

$$-2\Big(\frac{1}{2}\, h \otimes h + e \otimes f + f \otimes e\Big).$$

Exercise 6.7 (Strings that do not stop). This exercise departs from finite dimensionality, as Lecture 12 will. Let $\lambda \in \mathbb{C}$ and let M have basis $u_0, u_1, u_2, \dots$, with $u_{-1} = 0$ and

$$h u_k = (\lambda - 2k) u_k, \qquad f u_k = u_{k+1}, \qquad e u_k = k(\lambda - k + 1)\, u_{k-1}.$$

(i) Prove that M is an $\mathfrak{sl}_2$-module with highest vector u_0.

(ii) If $\lambda \notin \{0, 1, 2, \dots\}$, prove that M is irreducible.

(iii) If $\lambda = m \in \{0, 1, 2, \dots\}$, prove that $N = \mathrm{span}\{u_k : k \geq m+1\}$ is a submodule and $M/N \cong V_m$. Explain why the highest vector u_{m+1} of N does not contradict the highest-vector theorem of Lecture 6.